

Language Models are Few-Shot Learners

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Abstract

Recent work has demonstrated substantial gains on many NLP tasks and benchmarks by pre-training on a large corpus of text followed by fine-tuning on a specific task. While typically task-agnostic in architecture, this method still requires task-specific fine-tuning datasets of thousands or tens of thousands of examples. By contrast, humans can generally perform a new language task from only a few examples or from simple instructions. Here we show that scaling up language models greatly improves task-agnostic, few-shot performance. We train GPT-3, an autoregressive language model with 175 billion parameters, and test its performance in the few-shot setting.

1. Introduction

Recent years have featured a trend towards pre-trained language representations in NLP systems, applied in increasingly flexible and task-agnostic ways for downstream transfer. A limitation of the fine-tuning approach is the need for a large dataset of labeled examples for every new task, which limits the applicability of language models. In this paper, we test this hypothesis by training a 175 billion parameter autoregressive language model and measuring its few-shot learning abilities.

2. Approach

Our basic pre-training approach, including model, data, and training, follows the process described in prior GPT-2 work, with relatively straightforward scaling up of the model size, dataset size and diversity, and length of training. We focus on zero-shot, one-shot, and few-shot settings, in which the model is given a natural language description of the task and, optionally, a few demonstrations, but no gradient updates are performed.

2.1 Model and Architectures. We use the same model and architecture as GPT-2, including the modified initialization, pre-normalization, and reversible tokenization described therein, with the exception that we use alternating dense and locally banded sparse attention patterns in the layers of the transformer. We trained 8 different sizes of model, ranging from 125 million to 175 billion parameters, with the largest called GPT-3.

3. Results

On many NLP datasets, GPT-3 achieves promising results in the zero-shot and one-shot settings, and in the few-shot setting is sometimes competitive with or even occasionally surpasses state-of-the-art fine-tuned systems. We also find that GPT-3 can generate samples of news articles which human evaluators have difficulty distinguishing from human-written articles.

6. Limitations

Despite the strong quantitative and qualitative improvements of GPT-3, it still has notable weaknesses in text synthesis and several NLP tasks. On text synthesis, although the overall quality is high, GPT-3 samples still sometimes repeat themselves semantically at the document level, lose coherence over long

passages, and occasionally contradict themselves.

7. Conclusion

We presented GPT-3, a 175 billion parameter language model, and showed that it can achieve strong performance on many NLP tasks and benchmarks in zero-shot, one-shot, and few-shot settings, sometimes matching or exceeding the performance of fine-tuned systems, while also generating high-quality samples. These results suggest that very large language models may be an important ingredient toward general-purpose language systems.